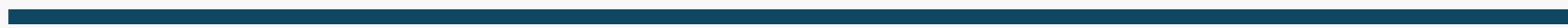


Prepared by Mamie Barnard, Nolan Steinbuch, & David Bori

Tidal & Wave Energy

Presenting the Two-Way Tidal Energy Model

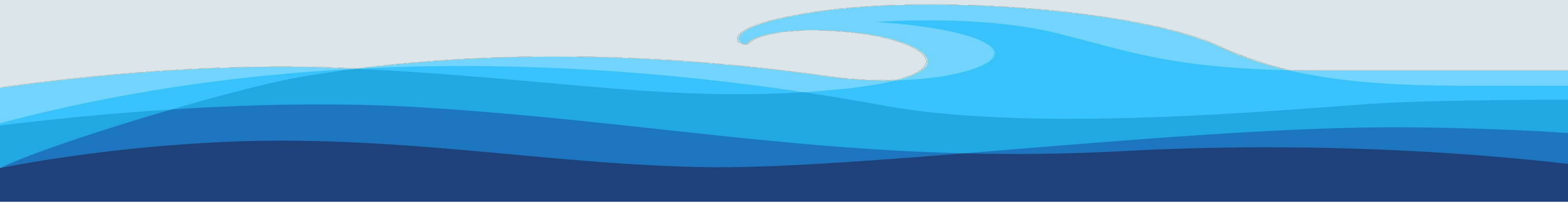
20 May 2025



Contents

1. Introduction to Tidal Energy
2. Research Paper Introduction
3. Methodology
 - Prototype
 - Time Scale Model
4. Experiments Overview
5. Findings
6. Limitations
7. Unanswered Questions & Possible Extensions

Introduction: What is Tidal Energy?



- **Renewable energy** which comes from the natural rise and fall of the ocean tide
 - Driven by the gravitational pull of sun/moon
 - Its energy can be **converted into electrical energy**
 - **Predictable** renewable energy sources

Tides vs Waves?

Tides

- Mainly caused by **gravitational pull**
- Rise and Fall of **sea level**
- Affects Wide **coastal regions**

Waves

- Mainly caused by **wind**
- Purely **surface motions** (ripples and swells)
- **Local**

Define Tidal/Wave Energy

- **Tidal energy** is the **energy obtained from the movement of tides** and is harnessed to produce electricity **using turbines or other energy conversion systems**. It can be categorized into:
 - **Tidal Stream Energy**
 - Harnesses the kinetic energy of moving water
 - **Tidal Range Energy**
 - Uses the potential energy from the height difference between high and low tides
 - **Dynamic Tidal Power**
 - A theoretical method using long dams to create head differences via phase shifts

Technical Side (Equations)

Tidal Stream Systems

- **Kinetic approach**
 - These are analogous to wind turbines and rely on the kinetic energy of moving water

- Governed by equation:

$$P = \frac{1}{2} \rho A v^3 C_P$$

Tidal Range Systems

- **Potential Energy Approach**
 - Uses change in tide height, (Tidal range) to generate energy
- Governed by equations:

$$E = \frac{1}{2} \rho g A h^2; P = \frac{E}{T}$$

Dynamic Tidal Power

- A **Theoretical Method** to Create head differences via phase shifts
- Governed by equations:

$$\Delta h = h_0 \sin\left(\frac{\omega L}{c}\right); P = \eta \rho g Q \Delta h$$

Different Applications of Tidal/Wave Energy

Tidal Energy can be used for a multitude of purposes:

- **Electricity Generation**

- The main application using turbines, but this can be used for several specific cases

- 1. Power for remote or coastal communities**

- 2. Desalination**

- Using seawater to produce freshwater

- 3. Hydrogen production**

- Using excess energy for electrolysis

- 4. Buoy power**

- Wave powered buoys for navigation or monitoring

- 5. Marine research**

- Powering Oceanographic instruments and tools

- 6. Offshore aquaculture**

- Powering fish farms or sensors

How do we harness it?



There are several methods of harnessing tidal energy:

- **Tidal Stream Turbines**
 - Turbines placed under water in tidal currents
- **Tidal Barrage**
 - A dam across estuaries trap water at high tide, then release it through turbines
- **Tidal Lagoons**
 - Artificial enclosures which work similar to barrages, but can be built offshore
- **Dynamic Tidal Power**
 - Long dam structures which create a pressure difference from tidal phase shifts
 - This approach is still theoretical



What are the current challenges?



1. High Maintenance and Operational Cost

- Still in the early stages of commercialization
- Corrosion and biofouling can result in high maintenance costs

2. Impact on Local Marine Life

- Turbines could cause harm to marine animals
- Large installations could disrupt migratory paths

3. Geographical Limitations

- Limited locations for implementation



Diagrams and Pictures:

Figure 1: Tidal Turbine Diagram

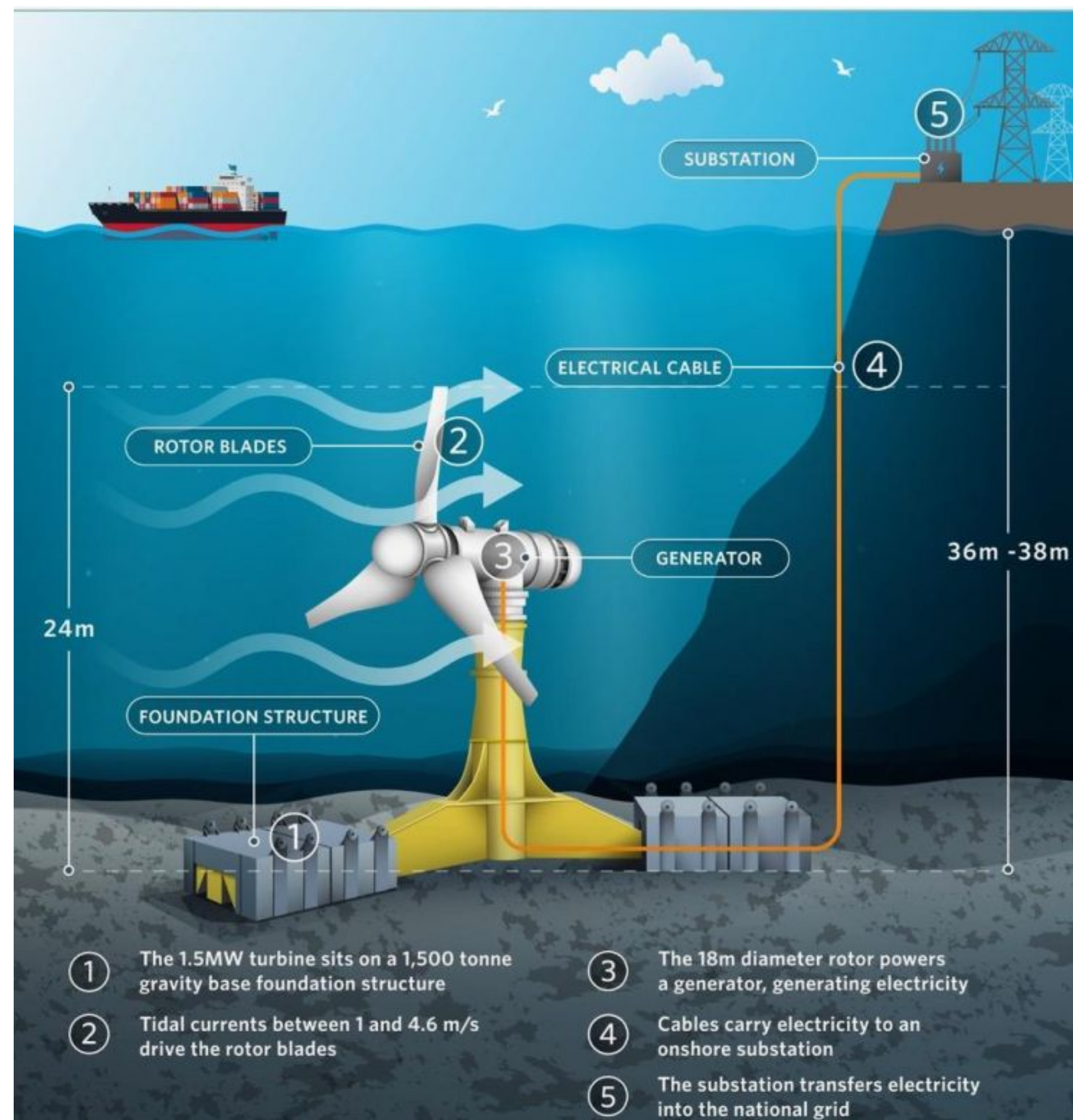
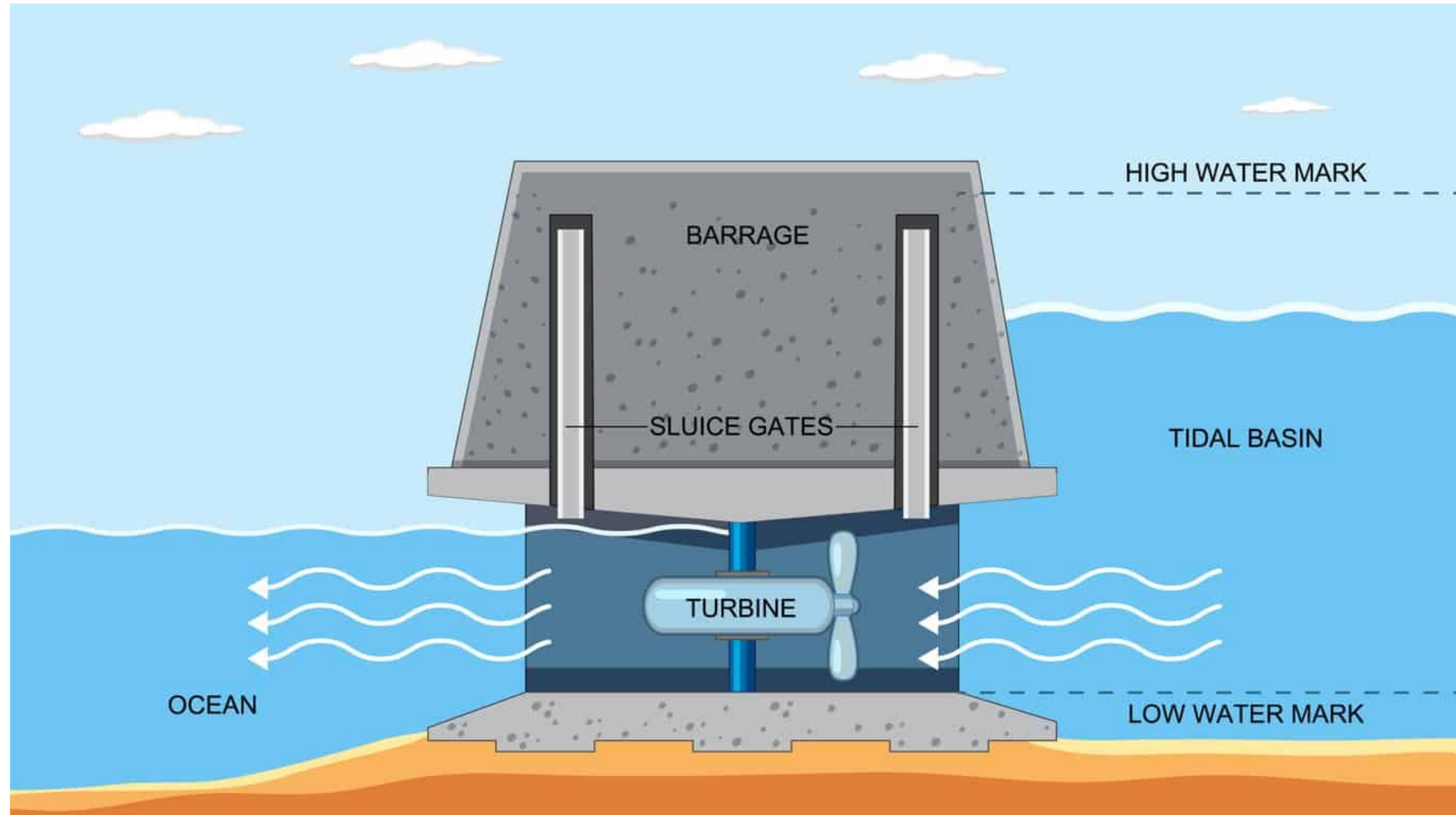


Figure 1: Renewables Advice. "How Does Tidal Energy Work Step by Step: A Closer Look." *Renewables Advice*, 2025, <https://renewablesadvice.com/energy/tidal-energy/>.

Figure 2: Tidal Barrage Diagram



Existing Tidal Power Plants

1. La Rance – France

- First large-scale tidal power station, 1966
- Six sluice gates, 720 m embankment
- Installed capacity of 240 MW
- Produces 480 GWh annually



1. Sihwa Lake Tidal Power Station – South Korea

- Largest tidal barrage currently in use
- Runs only when there is a flood, works to keep the surrounding infrastructure safe
- Installed capacity of 254 MW
- Produces 552 GWh annually



1. Jiangxia Experimental Tidal Power Plant – China

- Installed capacity of 4.1 MW
- Produces 6.5 GWh annually



Existing Tidal Power Plants

1. Kislaya Guba – Russia

- Only tidal plant in the Arctic, located in a fjord
- Shut down for 10 years until 2024
- Installed capacity of 1.7 MW

1. Haishan Tidal Power Plant – China

- Only operating linked-basin tidal scheme
- Includes pumping to increase storage capacity
- Installed capacity of 150 kW

1. Annapolis Royal Generating Station – Canada

- Single turbine that runs on ebb (outward flow) current
- Shut down in 2019



Research Paper Introduction

Evaluation of Tidal Energy Potential Using a Two-Way Energy Model, A. Aliffathur Rusvan, et al.

- Research and experimentation built around optimizing two-way tidal energy output for Indonesia
 - Introduces tidal energy as a technology, reviews current literature and operating tidal energy facilities
-

Barrage Tidal Basin

Evaluation of Tidal Energy Potential Using a Two-Way Energy Model, A. Aliffathur Rusvan, et al.

- **Basin and turbine** designed to output energy during the **flood (input)** and **ebb (output)** of tide water
- **Significant parameters:** time duration of flow, potential head difference

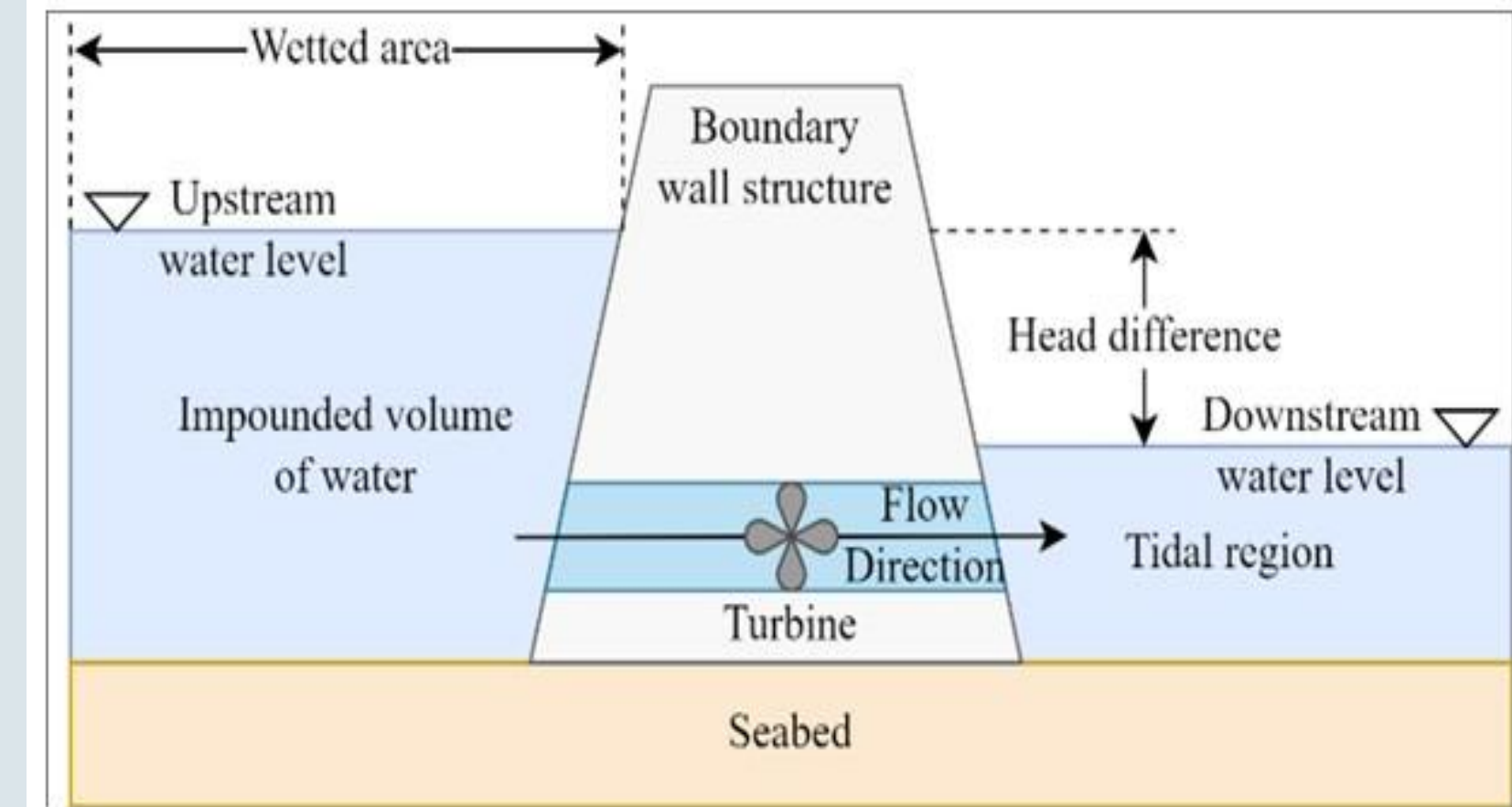


Figure 3: Two-Way Tidal Basin Schematic

Methodology Overview

The Prototype:

- Designed a functional tidal-range model
- Utilize three distinct variables to optimize and test model at a lower scale

Semidiurnal Tidal Time Scaled Model:

- Used prototype to test and verify theories in real time about model
- Drew conclusions and offered more perspectives into functionality of initial model

The Prototype

Layout:

- Initial tests varied the **land pond area**, dissimilarity of **tidal height difference**, and **flow hole diameters**
- Tidal Range system
- Water flows from **higher reservoir to lower**, potentially spinning a turbine or driving a pump

Variables:

- **Variance of land pond area** affects **generation time** (wider pond means more generation time)
- **Flow holes** affect **total mass flow rate** of the model (bigger holes means more potential to generate energy but less control of speed)
- **Tidal height difference** (head) **affects gravitational force** driving water through turbine (greater head means more potential energy)

Figures 4 & 5: Experimental Setup

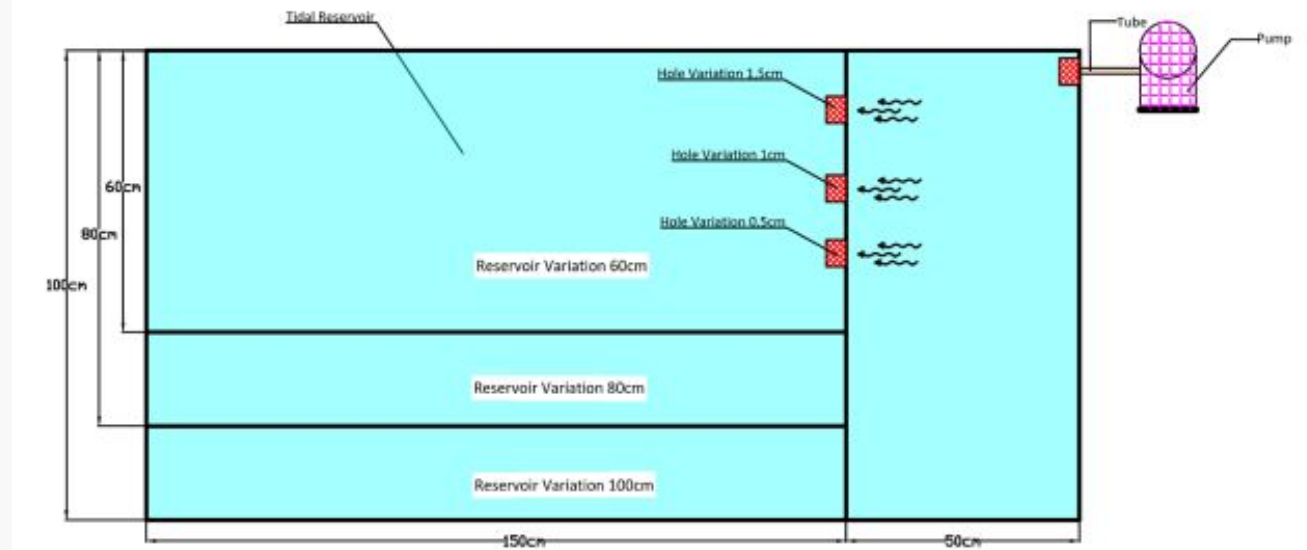


Figure 12. Top view of the 2D design of the tidal energy capture prototype model

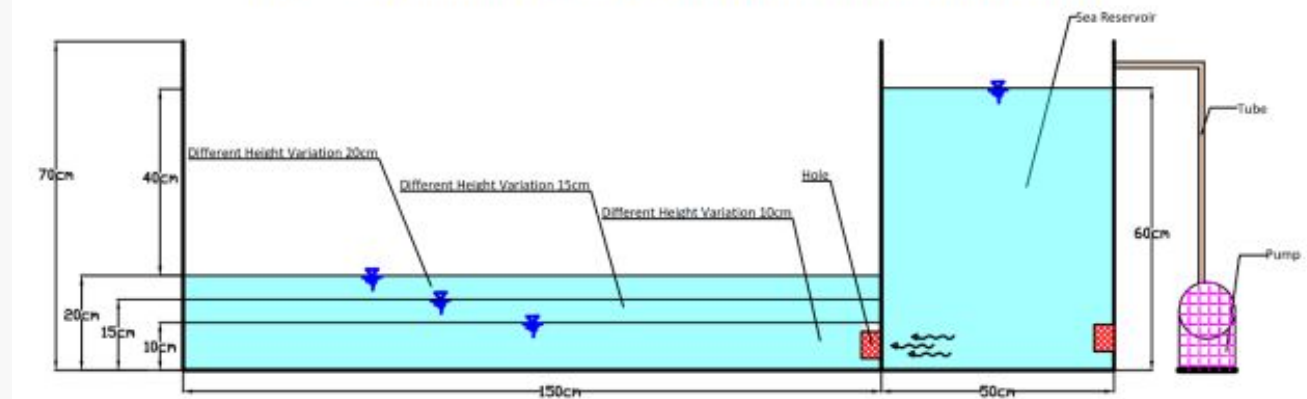


Figure 13. Longitudinal section of the 2D design of the tidal energy capture prototype model

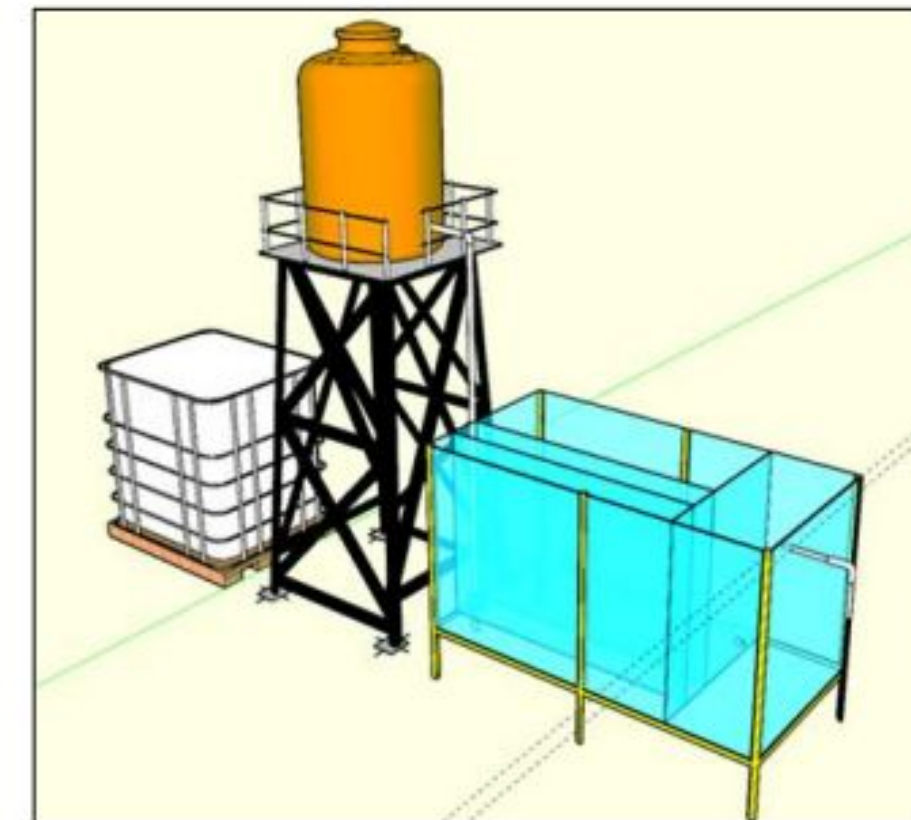


Figure 14. 3D design of the tidal energy capture prototype model

Time Tidal Model

Model:

- Smaller scale model that can provide a preliminary idea on the duration of the cycles
- Semidiurnal cycle: 2 high tides and two low tides (i.e 6 hour cycle in real world time)
- Model scaled down (1:10 ratio) and simulated with different heads
- Greater head results in shorter cycle due to more force driving the pump/turbine

Significance:

- Helps team test full tidal cycles faster (instead of waiting for 24 hours)
- Helps insure that the physics of flow and turbine behavior accurately match real-world tidal patterns

Figures 6 & 7: Time Tidal Model

Table 2. Time Scale in the Model Research

Scale	Time
Scale 1:1	6 hours
Scale 1:10	6830.52 seconds $\approx$ 1 hour 53.84 minutes
Height difference 10 cm	5692.1 seconds $\approx$ 1 hour 34.87 minutes
Height difference 15 cm	5122.89 seconds $\approx$ 1 hour 25.38 minutes
Height difference 20 cm	4553.68 seconds $\approx$ 1 hour 15.89 minutes

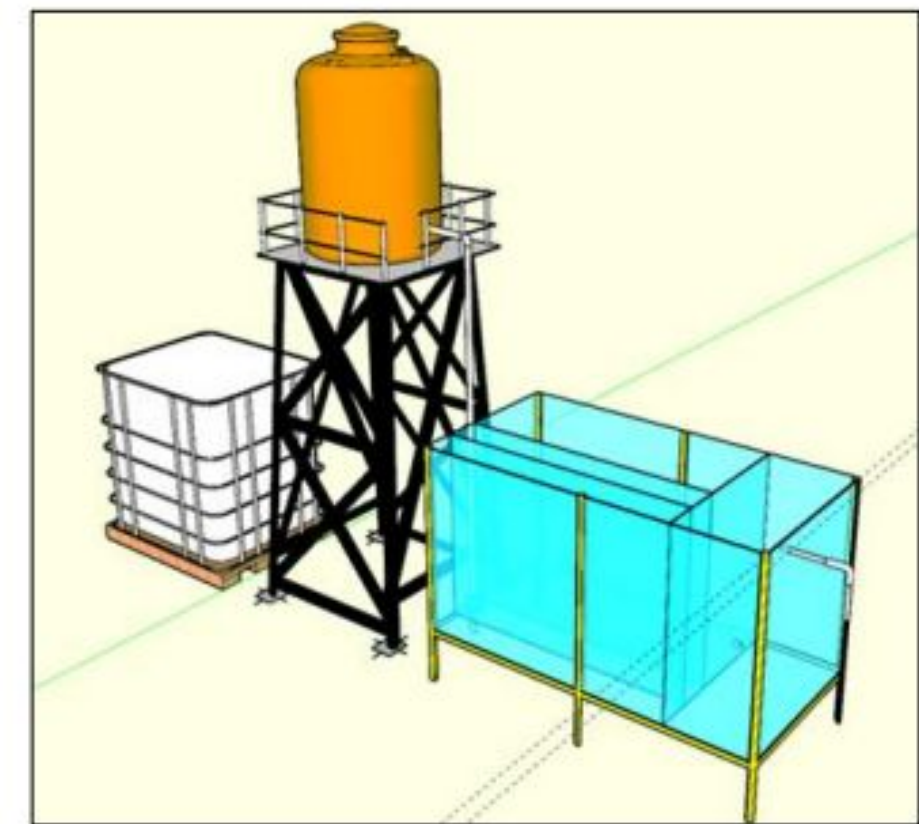


Figure 14. 3D design of the tidal energy capture prototype model

Figures 10 & 11: Experiment 1 Results

Experiment 1

Conditions:

- **Pond width** fixed at **100 cm**
- Constant **hole size fixed at 1.5 cm** diameter
- **Tidal height difference** (head) **varied** from 10 to 20 cm

Results:

- **10 cm difference** (purple line) **takes the longest** (6 minutes) to reach the peak size but has the highest peak
- **15 cm difference** and **20 cm difference peak much sooner** but overall water level is lower

Reasoning:

- Lower head results in more stability → rises to a higher peak
- Greater head produces shorter cycle (expected)

Table 3. Relationship between the water level in the land pond and time with a hole diameter of 1.5 cm					
Water Level in the Land Pond (cm) height difference 10 cm	Water Level in the Land Pond (cm) height difference 15 cm	Water Level in the Land Pond (cm) height difference 20 cm	Time (seconds) height difference 10 cm	Time (seconds) height difference 15 cm	Time (seconds) height difference 20 cm
0	0	0	0	0	0
5	5	5	563.08	499.8	414.85
10	10	10	499.92	466.99	401.78
15	15	15	521.67	504.82	420.79
20	20	20	530.11	507.31	439.53
25	25	25	530.84	493.76	453.46
30	30	30	538.92	508.24	453.53
35	35	35	539.34	491.93	467.77
40	40	40	532.75	519.16	458.19
45	45	45	543.29	505.61	466.13
50	50	50	537.08	515.49	476.86
53.1	51.3	51.6	355.1	109.78	100.79

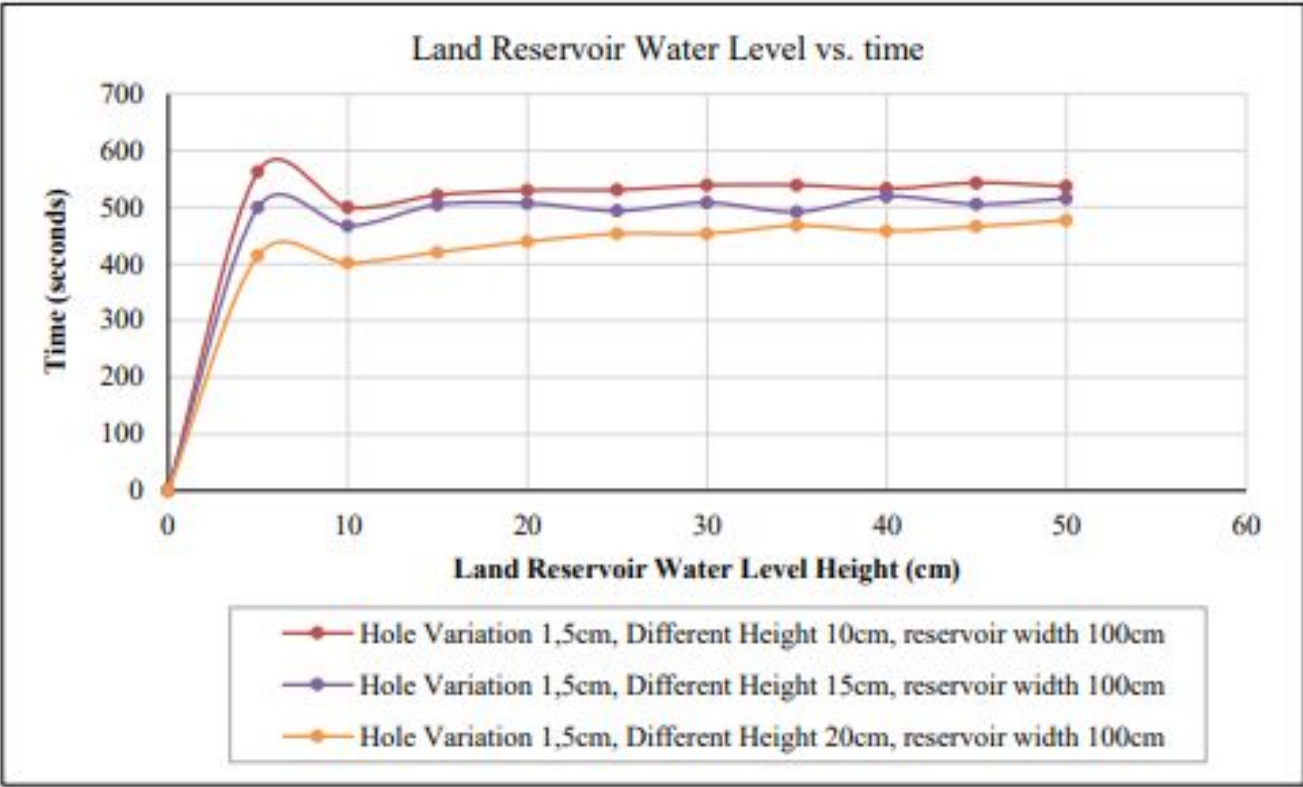


Figure 17. The relationship between water level in the land pond and time at a tidal height difference of 10 cm

Experiment 3

Conditions:

- **Pond width** fixed at **100 cm**
- Constant **hole size** fixed at **0.5 cm** diameter
- **Tidal height difference** (head) **varied** from 10 to 20 cm

Results:

- Very little variation in final height
- **10 cm head still resulted in highest peak**
- Different cases each had **similar cycle time**
- Compared to experiment 1 and 2, **peak heights were much lower in this experiment**

Reasoning:

- Lower hole diameter causes lower mass flow rate (i.e lower peak height but a shorter duration)
- Mass flow rate of water is a large limiting factor in amount of height produced

Figures 12 & 13: Experiment 3 Results

Table 5. Relationship between the water level in the land pond and time with a hole diameter of 0.5 cm					
Water Level in the Land Pond (cm) height difference 10 cm	Water Level in the Land Pond (cm) height difference 15 cm	Water Level in the Land Pond (cm) height difference 20 cm	Time (seconds) height difference 10 cm	Time (seconds) height difference 15 cm	Time (seconds) height difference 20 cm
0	0	0	0	0	0
5	5	5	1893.18	1713.24	1489.31
10	10	10	1685.52	1317.68	1137.29
15	15	15	1101.12	1160.14	1080.48
19.6	19.5	19.2	1012.28	931.83	846.6

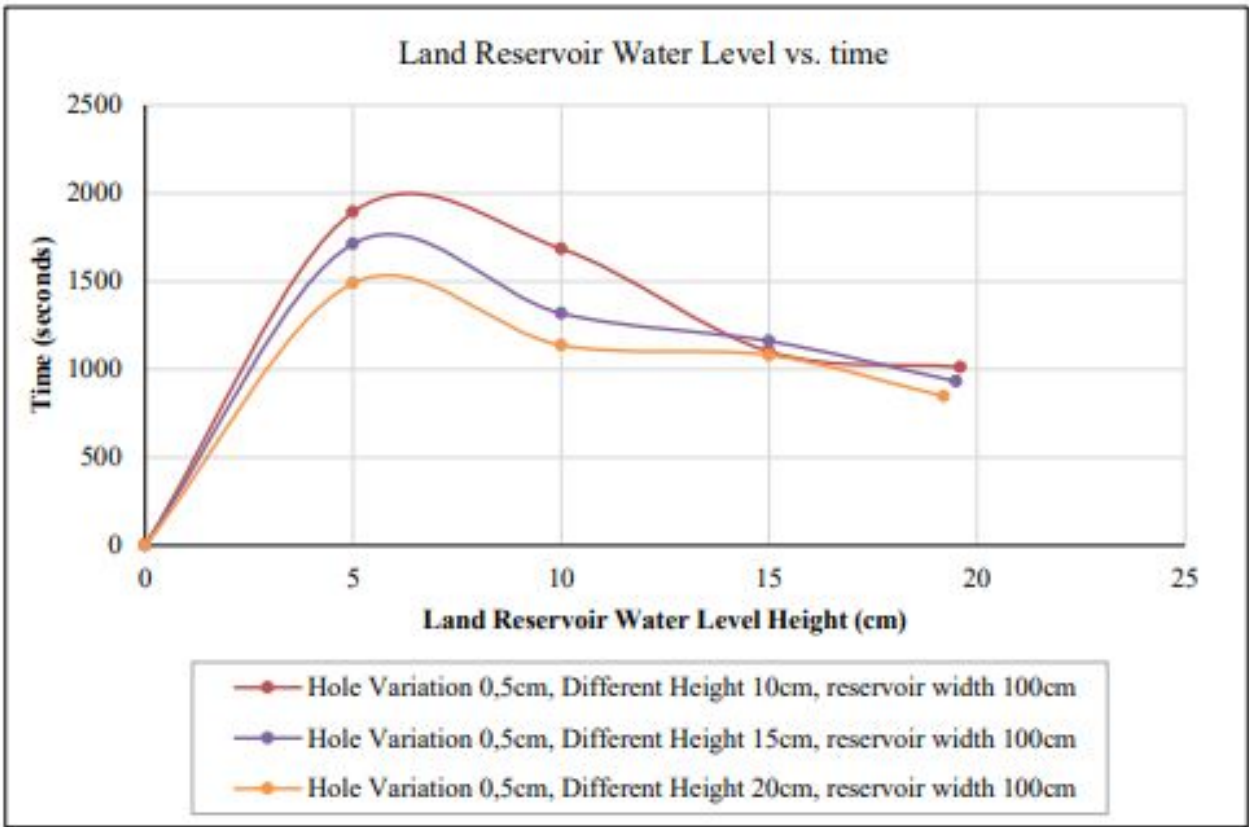


Figure 19. The relationship between water level in the land pond and time at a tidal height difference of 20 cm

Energy Calculation

Governing Equation:

- $E = \frac{1}{2} * (\eta * \rho * g * A * H^2)$
- Depends on gravity, surface area of tidal basin, efficiency, density of fluid (water), and head

Results:

- Smaller hole size corresponds to a higher energy generated

Reasoning:

- With a smaller hole, water enters the land pond **more slowly**, meaning the **head difference (height)** between sea and pond lasts longer.
- This extended pressure duration allows **more consistent energy production** over time.
- With smaller holes, you have **sustained flow**, so even if it's slower, it **lasts longer**, adding up to more energy over time.

Figures 13 & 14: Energy Calculation Results

Table 8. Relationship between energy and 10 cm height difference							
Length of Land Pond (cm)	Width of Land Pond (cm)	Height Difference at 1.5 cm hole	Height Difference at 1 cm hole	Height Difference at 0.5 cm hole	Energy at 1.5 cm hole (kWh)	Energy at 1 cm hole (kWh)	Energy at 0.5 cm hole (kWh)
150	100	10	10	10	146.6	146.6	146.6
150	100	11.6	18.4	27.5	197.3	496.4	1108
150	100	10.2	20.8	36.6	152.5	634.4	1964
150	100	9.6	22.2	39	135.1	722.7	2230
150	100	9.3	22.5	40.4	126.8	742.3	2393
150	100	9.2	21.8	-	124.1	696.8	-
150	100	9.1	21.4	-	242.8	1343	-
150	100	9.8	21.1	-	281.6	1305	-
150	100	9	20.8	-	237.5	1268	-
150	100	8.9	-	-	232.3	-	-
150	100	8.9	-	-	232.3	-	-
150	100	6.9	-	-	139.6	-	-

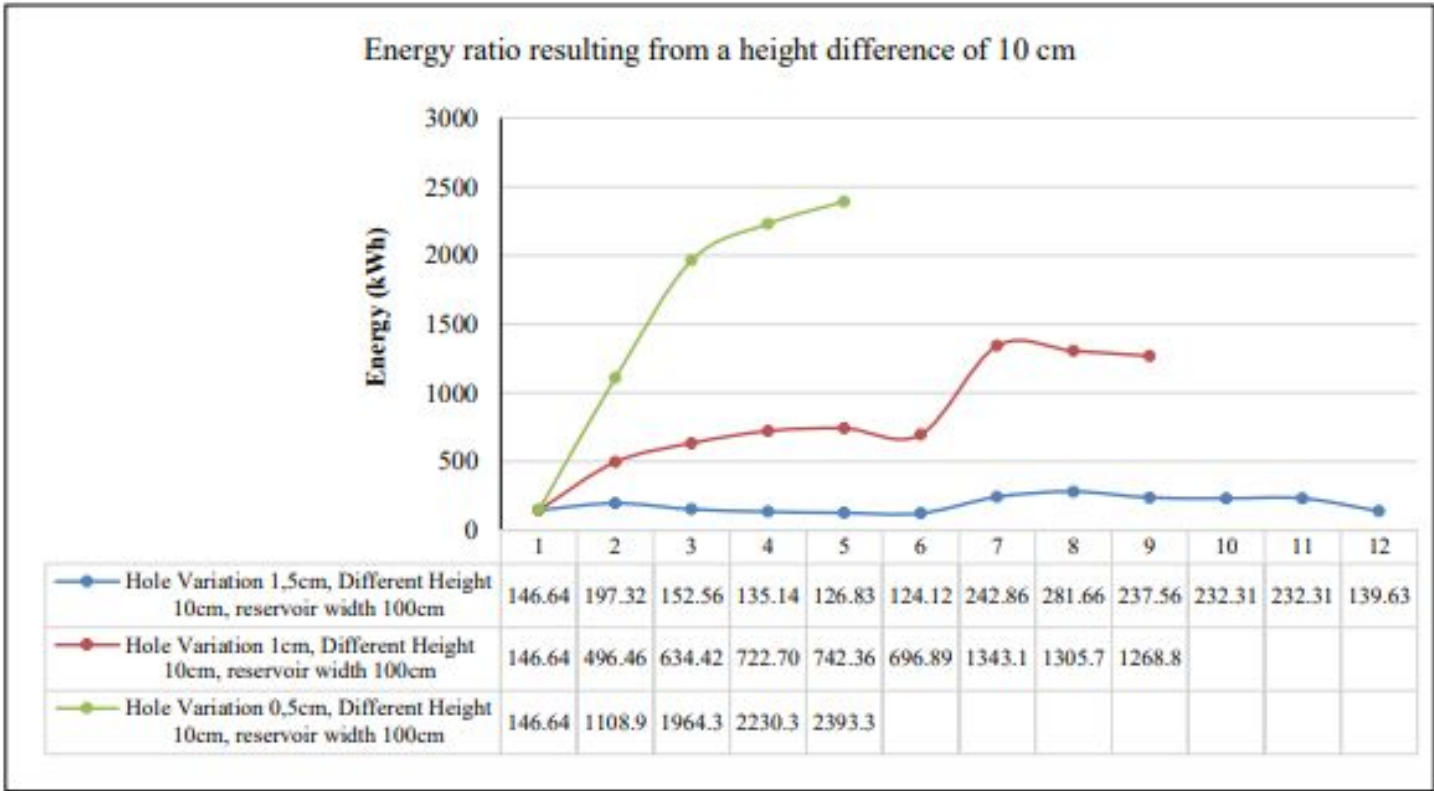


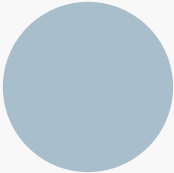
Figure 26. Energy produced at a tidal height difference of 10 cm

Findings

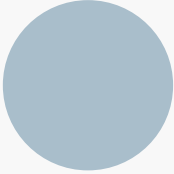
Key Takeaway: Although these results were found for a specific location in Indonesia, the identified relationships between basin dimensions, flow hole diameter, flow hole height, and energy output are crucial for the commercialization of two-way tidal basin designs.

- **Potential Head Height Difference:** Pond dimensions of 150×100 cm and 150×80 cm showed the highest potential for height differences
- **Time Duration of Water Flow:** For 1.5 cm hole diameter, pond dimensions of 150×60 cm and a height difference of 10 cm required the most time for the water basin to reach peak height
- **Energy Output:** For 1.5 cm flow hole diameter, max energy output was 281.84 kWh. For 1 cm flow hole diameter, max of 1803.78 kWh. For 0.5 cm flow hole diameter, max of 2623.8 kWh.

Limitations



Semidiurnal Model: Experiments run exclusively with a semidiurnal model. While many shores experience semidiurnal tides, this is not true for every location and could impact the results found.



Scale Model: Results found using a scale model, rather than field testing. This is due largely to the lack of large scale implementation of this technology.



Implementation Challenges: Limited discussion into the economic or technological challenges that Indonesia would face

Unanswered Questions & Possible Extensions

- What would be the **economic feasibility** of implementing a barrage tidal basin in the location described?
- How specifically would the implementation of such a system **impact the local marine life**?
- How can these results best be utilized in the broader research goal of **tidal energy converter commercialization**?



Questions?



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Figure References

Fig 1-2: Renewables Advice. “How Does Tidal Energy Work Step by Step: A Closer Look.”

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Figs 3-14: Rusvan, A. Aliffathur, et al. “Evaluation of Tidal Energy Potential Using a Two-Way Tidal Energy Model.” *Civil Engineering Journal*, vol. 10, no. 9, 1 Sept. 2024, pp. 3011–3033,

<https://doi.org/10.28991/cej-2024-010-09-016>.